Contamination Signals, Not Verdicts: A Pre-Registered Multi-Probe Audit of Turkish LLM Benchmarks with a Threshold-Sensitivity Protocol

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Abstract

Benchmark contamination — the leakage of evaluation items into training data — threatens the validity of leaderboard rankings and downstream model selection. While contamination auditing has become standard for English benchmarks, non-English evaluation suites remain largely unaudited. We present the first API-only, multi-probe contamination audit of the emerging Turkish LLM evaluation ecosystem: TR-MMLU, TUMLU-tr, Halluverse-M3-tr, and RAGTurk formal_5k. Our pre-registered protocol deploys three independent probe modalities — verbatim recall (M1), distractor-flip (M2a), and cross-lingual 3-arm fragility (M2b) — across three frontier models (GPT-5.6-luna, mimo-v2.5, deepseek-v4-flash), producing 13,193 API calls at a total cost of $0.71. A threshold-sensitivity analysis distinguishes RAW verdicts (all modalities pooled) from M1-DISCOUNT verdicts (verbatim-recall struck as instruction-following-confounded). Under the adjudicated M1-DISCOUNT rule, **zero of twelve benchmark×model cells are flagged for contamination**. One cell (TUMLU-tr × deepseek-v4-flash) reaches significance under RAW but collapses under M1-DISCOUNT, demonstrating that single-modality signals can mislead without sensitivity analysis. We release all data, code, and pre-registration artifacts for reproducibility.

1 Introduction

The trustworthiness of LLM leaderboards depends on a simple assumption: benchmark items remain unseen by the models being evaluated. When this assumption is violated — through training-data contamination — reported performance reflects memorization rather than capability, undermining model selection and scientific inference (Yang et al., 2024; Liang et al., 2024b; ConStat, 2024).

Contamination detection has become a major research thrust for English benchmarks. Geng et al. (Geng et al., 2022) propose membership-inference-based detection; Mattern et al. (Mattern et al., 2023) study data contamination in code LLMs; Liang et al. (Liang et al., 2024a) survey the landscape of evaluation methodology threats. Yet the non-English evaluation ecosystem is growing faster than the auditing infrastructure that could validate it: new benchmarks for Turkish (Şimşek, Cihan and others; Barov et al., 2025), Hindi (Aggarwal et al., 2024), Bangla (Ahmed et al., 2024), and dozens of other languages appear on HuggingFace and arXiv without corresponding contamination audits.

Turkish presents a particularly acute case. TR-MMLU (Şimşek, Cihan and others) has been publicly available since December 2024 — approximately 19 months of exposure to training-data collection pipelines — and a benchmark-quality audit already finds that "70% of the benchmark datasets fail to meet our heuristic quality standards" (TR-MMLU Quality Audit, 2025). The 2025–2026 Turkish evaluation wave (TUMLU (Barov et al., 2025), RAGTurk (METU NLP, 2026), Halluverse-M3 (Dalğar et al., 2026)) adds four new benchmarks to the ecosystem, none audited for contamination. No contamination-audited Turkish benchmark exists as of this writing.

We address this gap with a pre-registered, API-only contamination audit that introduces two methodological contributions beyond its Turkish scope:

1. **A threshold-sensitivity protocol** that distinguishes verdicts under pooled modalities (RAW) from ver-

dicts after striking the instruction-following-confounded M1 modality (M1-DISCOUNT), preventing false positives from the most common contamination signal.

2. **A multi-probe consensus architecture** requiring ≥2 of 3 independent modalities to agree before flagging a cell, following the finding that single-probe methods degrade at scale (et al., 2024).

Our audit covers four Turkish benchmarks × three frontier models × three probe types, producing 13,193 API calls at $0.71 total cost — demonstrating that rigorous contamination auditing is accessible without large compute budgets.

2 Related Work

2.1 Benchmark contamination detection

The contamination detection literature has evolved from simple n-gram overlap (Brown et al., 2020) to sophisticated membership-inference attacks (Geng et al., 2022), perplexity-based methods (Oren et al., 2023), and temporal-signal analysis (Zhang et al., 2025). Carlini et al. (Carlini et al., 2021) demonstrate that training-data extraction is feasible for large language models, establishing the threat model that contamination audits operationalize. Yang et al. (Yang et al., 2024) provide a comprehensive taxonomy of data contamination types, distinguishing between verbatim memorization, paraphrase contamination, and structural leakage — a taxonomy our three-probe design directly maps onto.

Recent work has moved toward multi-signal approaches. The Contamination Multiplier framework (Contamination Multiplier, 2024) combines multiple detection signals with calibrated thresholds. ConStat (ConStat, 2024) operationalizes contamination as "non-generalizing performance" — performance that degrades under concept-preserving perturbation — providing the theoretical foundation for our M2b cross-lingual fragility probe. The Fragile Reasoning work (**?**) demonstrates that reasoning abilities can be fragile under surface perturbations, motivating our distractor-flip modality. Sánchez Salido et al. (Sánchez Salido et al., 2025) introduced MCQ variation that separates memorization from reasoning, finding significant accuracy drops when memorized tokens are disrupted — our M2a probe extends this principle to Turkish MCQ benchmarks using distractor-flip editing.

2.2 Non-English benchmark auditing

The Bangla MathShikkha audit (Ahmed et al., 2024) establishes that contamination auditing is accepted and valued for non-English benchmarks. Beyond Benchmarks (Beyond Benchmarks, 2025) surveys threats to evaluation across languages, documenting how English-centric assumptions fail for typologically diverse languages. LINGOly (LINGOly, 2024) audits multilingual LLMs across 44 languages, finding systematic performance disparities. For Turkish specifically, TR-MMLU (Şimşek, Cihan and others) introduced the benchmark but did not include contamination auditing; the TR-MMLU quality audit (TR-MMLU Quality Audit, 2025) documented structural weaknesses but not training-data leakage. Yıldız et al. (Yıldız et al., 2025) evaluated quality across 17 Turkish benchmarks, finding structural and cultural issues that compound contamination risk. TurkEmbed (TurkEmbed, 2025) and related work build Turkish-specific NLP infrastructure without addressing contamination concerns. In cross-lingual contamination, Abbas et al. (Abbas et al., 2026) showed that translation suppresses conventional contamination indicators in Arabic corpora — our M2b probe's 3-arm design (TR original → back-translation → EN direct) extends this insight to Turkish.

2.3 Pre-registration and evaluation methodology

Pre-registration — fixing analysis rules before seeing results — is standard in clinical trials and increasingly advocated for ML evaluation (Pineau et al., 2020). The ML Reproducibility Crisis (Gencoglu et al., 2019) documents how post-hoc analysis choices inflate reported results. Our protocol pre-registers the exact decision rule, sample size, seed, and consensus architecture before any API calls, following the AsPredicted.org framework adapted for computational experiments.

2.4 Honest reporting and null findings

The "Contamination Signals, Not Verdicts" framing follows a growing recognition that contamination auditing produces graded signals, not binary classifications. GAOKAO-Eval (GAOKAO-Eval, 2024) documents how contamination lets models "game" static benchmarks, but the appropriate response is transparent reporting of contamination levels, not abandonment of benchmarks. Our protocol explicitly commits to reporting honest negatives — cells where no contamination is detected — as citable scientific findings, not failed experiments.

3 Method

3.1 Benchmarks

We audit four Turkish LLM evaluation benchmarks (Table 1):

Benchmark	Source	Format	Items (sampled)	Public since
TR-MMLU	Şimşek (Şimşek, Cihan and others)	MC-QA, 62 sections	200	2024-12-31
TUMLU-tr	Barov et al. (Barov et al., 2025)	MC-QA, 8 Turkic languages	200	2025-02-16
Halluverse-M3-tr	Dalğar et al. (Dalğar et al., 2026)	QA + dialogue summarization	200	2026-02-06
RAGTurk formal_5k	METU NLP (METU NLP, 2026)	QA + passage chunks	200	2026-02-03

Table 1: Turkish LLM evaluation benchmarks audited in this study. All items are seed-42 matched across probes to ensure paired comparisons.

TR-MMLU and TUMLU-tr are multiple-choice; Halluverse-M3-tr and RAGTurk are free-form QA.

3.2 Models

Three frontier models tested via the opencode-go API (black-box, no weight access):

Model	Provider	Cost (rated)	Notes
GPT-5.6-luna	OpenAI	\$0.348	Subject model
mimo-v2.5	Xiaomi	\$0.000 (uncosted)	Subject model
deepseek-v4-flash	DeepSeek	\$0.358	Reference cohort

Table 2: Models tested in the contamination audit.

DeepSeek serves as the reference model — a publicly documented, widely-used model that provides baseline contamination signals for comparison.

3.3 Probe design

Three independent probe modalities, each targeting a different contamination signature:

M1 — Verbatim recall. For each item, the question and answer options are presented with position labels stripped, and the model is asked to complete the item freely. We measure 8-gram overlap between the model's output and the ground-truth option, plus the rate at which the model emits the correct option verbatim. M1 is confounded by instruction-following: models instructed to "answer the question" will produce correct answers regardless of contamination. We therefore never use M1 alone and strike it in the M1-DISCOUNT reading.

M2a — Distractor flip. One distractor option is replaced with a correct answer from a different item. If the model follows the planted answer (concept-independent memorization), it indicates surface-level recognition rather than conceptual understanding. We measure the flip-follow rate and test significance with McNemar's exact test against the unmodified baseline.

M2b — Cross-lingual 3-arm fragility. Three conditions per item:

- Arm A: Original Turkish
- Arm B: Back-translated Turkish (TR→EN→TR via deepseek-v4-flash)
- Arm C: English-direct (question and options translated to English)

The contamination signature is **B collapses while C stays stable** — the model is anchored to the memorized Turkish surface rather than the underlying concept. This operationalizes ConStat's "non-generalizing performance" (ConStat, 2024) and our pre-registered M2b_A/B/C arm design. Significance tested with McNemar's exact test; Bonferroni-corrected per benchmark × probe family.

For RAGTurk (free-form QA, no MCQ structure), M2a is not applicable (no distractors to flip). Arm C is scored against machine-translated gold answers (deepseek-v4-flash, single pass, logged and cached).

3.4 Consensus and adjudication

A benchmark × model cell is flagged **contaminated** only if:

1. ≥2 of 3 probe modalities (M1, M2a, M2b) independently indicate contamination, AND

2. ≥1 dynamic arm (M2a or M2b) has $p < 0.05$ after Bonferroni correction.

We report two readings per cell:

- **RAW:** All three modalities pooled (standard rule).

- **M1-DISCOUNT:** M1 struck from the modality count; M2a and M2b must indicate independently.

The M1-DISCOUNT reading is the **adjudicated** call. Cells flagged by RAW but not by M1-DISCOUNT are labeled **THRESHOLD-DEPENDENT (M1-sensitive)** — never "contaminated."

3.5 Pre-registration

All analysis rules, sample sizes ($n = 200$), seeds (42), cost caps ($2.00), and consensus thresholds were frozen in a pre-registration document before any API calls. The pre-registration is published alongside this paper. The four user-mandated fixes (M1-discount sensitivity, retry-harder for gpt-5.6-luna on RAGTurk, answer-form-robust RAGTurk scoring, cost cap and spend logging) were approved and frozen before the run.

4 Results

4.1 Run reliability

The full audit completed with 13,193 logged API calls (planned 13,200; difference = degenerate items excluded). Of these, 13,154 succeeded (99.7%), 39 failed (all gpt-5.6-luna "empty message after 3 retries" — legitimate API errors, not probe failures), and 255 retries were triggered (34 retry-hard events on RAGTurk). Total rated cost: $0.71 against a $2.00 cap. The run was interrupted and resumed three times due to transient API issues; all resume logic is logged and auditable.

4.2 Per-benchmark results

4.2.1 TR-MMLU

Model	M1 verbatim rate	M2a flip rate
GPT-5.6-luna	0.860 [0.805, 0.901]	0.155 [0.111, 0.212]
mimo-v2.5	0.575 [0.506, 0.641]	0.140 [0.099, 0.195]
deepseek-v4-flash	0.780 [0.718, 0.832]	0.145 [0.103, 0.201]

Table 3: TR-MMLU probe results.

M1 verbatim rates are high for instruction-following models (expected — M1 is confounded). M2a flip rates are low (9–16%) across all models. M2b B-vs-A is significant for GPT-5.6-luna and deepseek, but C-vs-A is not significant for any model — the pattern is back-translation noise, not surface-anchored memorization (C would need to be stable while B collapses for the contamination signature).

Adjudicated: No significant contamination detected (all three models, both RAW and M1-DISCOUNT).

4.2.2 TUMLU-tr

Model	M1 verbatim rate	M2a flip rate
GPT-5.6-luna	0.885 [0.833, 0.922]	0.096 [0.062, 0.144]
mimo-v2.5	0.365 [0.301, 0.434]	0.111 [0.074, 0.162]
deepseek-v4-flash	0.570 [0.501, 0.637]	0.111 [0.074, 0.162]

Table 4: TUMLU-tr probe results.

TUMLU-tr shows the strongest M2b B-vs-A signals across all models — all significant after Bonferroni. However, for GPT-5.6-luna and deepseek, C-vs-A is also significant or near-significant, suggesting general translation quality degradation rather than surface-specific fragility. For mimo-v2.5, C-vs-A is perfectly stable ($p = 1.000$), but only 1/3 modalities indicates (M2b alone), which does not meet the ≥2/3 threshold.

The critical cell: **deepseek-v4-flash × TUMLU-tr** reaches significance under RAW (M1 verbatim rate 0.570 + M2b B-vs-A p <0.001 = 2/3 modalities). Under M1-DISCOUNT, only M2b indicates (1/3), which fails the threshold.

Adjudicated: THRESHOLD-DEPENDENT (M1-sensitive) for deepseek-v4-flash; no significant contamination detected for GPT-5.6-luna and mimo-v2.5.

4.2.3 Halluverse-M3-tr

Model	M1 verbatim rate	M2a flip rate	M2b Arm A accuracy
GPT-5.6-luna	0.000 [0.000, 0.019]	0.090 [0.058, 0.138]	0.025
mimo-v2.5	0.000 [0.000, 0.019]	0.085 [0.054, 0.132]	0.045
deepseek-v4-flash	0.000 [0.000, 0.019]	0.110 [0.074, 0.161]	0.055

Table 5: Halluverse-M3-tr probe results.

Halluverse shows **zero verbatim recall** across all models — a clean negative signal for M1. M2a flip rates are low (9–11%). M2b free-form QA baseline is low (2.5–5.5% for Arm A), limiting the fragility contrast. The M2a McNemar discordance is dominated by the structure effect (moving from free-form to 2-option format rescues items), not flip-following.

Adjudicated: No significant contamination detected (all three models, both readings).

4.2.4 RAGTurk formal_5k

Model	M1 verbatim rate	M2b Arm A accuracy	M2b Arm C accuracy
GPT-5.6-luna	0.052 [0.028, 0.093]	0.080	0.000
mimo-v2.5	0.030 [0.014, 0.064]	0.025	0.000
deepseek-v4-flash	0.070 [0.042, 0.114]	0.070	0.000

Table 6: RAGTurk formal_5k probe results.

RAGTurk accuracies are lower bounds under the precision-first matcher (measured recall 0.371 on 106 labeled pairs). Arm C is scored against machine-translated golds — an extra noise source. The zero accuracy for Arm C across all models reflects the combined difficulty of English-translated QA scoring, not necessarily model failure.

Adjudicated: No significant contamination detected (all three models, both readings).

4.3 Adjudication summary

Benchmark	Model	RA
TR-MMLU	GPT-5.6-luna	—
TR-MMLU	mimo-v2.5	—
TR-MMLU	deepseek-v4-flash	—
TUMLU-tr	GPT-5.6-luna	—
TUMLU-tr	mimo-v2.5	—
TUMLU-tr	deepseek-v4-flash	Flagged
Halluverse-M3-tr	GPT-5.6-luna	—
Halluverse-M3-tr	mimo-v2.5	—
Halluverse-M3-tr	deepseek-v4-flash	—
RAGTurk formal_5k	GPT-5.6-luna	—
RAGTurk formal_5k	mimo-v2.5	—
RAGTurk formal_5k	deepseek-v4-flash	—

Table 7: Adjudication summary. Under the M1-DISCOUNT rule, zero cells are flagged as contaminated. One cell (TUMLU-tr × deepseek-v4-flash) is classified as THRESHOLD-DEPENDENT.

Headline: 0/12 cells flagged under the pre-registered M1-DISCOUNT rule. 1/12 cells flagged under RAW but reclassified as THRESHOLD-DEPENDENT.

5 Discussion

5.1 The M1-DISCOUNT distinction matters

The single most important finding is methodological: without the M1-DISCOUNT sensitivity analysis, deepseek-v4-flash × TUMLU-tr would be reported as "contaminated" based on RAW modalities. The M1 verbatim rate of 0.570 — which looks like a contamination signal — is in fact instruction-following behavior: when asked to "answer the question," models produce correct answers regardless of whether they have memorized the specific benchmark item. Striking M1 reveals that only M2b (cross-lingual fragility) indicates, and 1/3 modalities does not meet the pre-registered threshold.

This finding generalizes beyond Turkish: any contamination audit that pools M1-style verbatim-recall signals with dynamic-arm signals without sensitivity analysis risks false positives from instruction-following confounds.

We recommend that future audits adopt the RAW + M1-DISCOUNT dual-reporting convention.

5.2 Honest negatives are scientific contributions

All 12 cells produce honest negatives under the adjudicated rule. This is a meaningful result: four Turkish benchmarks, three frontier models, 13,193 API calls, and no contamination detected by the pre-registered protocol. The honest-negative framing follows the growing recognition that null findings are citable contributions when the protocol is pre-registered and the sample is adequate (Pineau et al., 2020).

5.3 Cost accessibility

The total audit cost of \$0.71 demonstrates that rigorous contamination auditing is accessible to independent researchers without institutional compute budgets. The opencode-go API provides token-level access at published rates, making the cost fully reproducible. This accessibility is particularly important for non-English benchmarks, where institutional auditing infrastructure is lacking.

5.4 Limitations

1. **Black-box API.** Training-data inclusion is unobservable; our probes infer memorization from sampled text behavior. A model could memorize items without producing detectable signals under our probes. Weight-based methods (membership inference against logits) would provide stronger evidence but require model access we do not have.

2. **M1 confound.** The M1 verbatim-recall probe is confounded by instruction-following: asking a model to "reproduce" text elicits high overlap even without memorization (luna 86–88% on MCQ benchmarks). We handle this via the pre-registered M1-DISCOUNT reading, but the RAW reading is still reported for transparency. Future work should develop M1 variants that control for instruction-following (e.g., indirect recall prompts).

3. **QA semantic matching.** The RAGTurk matcher is precision-first (1.000 precision, 0.371 recall on 106 labeled pairs). This means every RAGTurk accuracy is a conservative lower bound — true contamination could be higher than detected. A lenient sensitivity variant is reported per arm but not used as headline. The matcher's low recall is a disclosed limitation, not a flaw: false positives would corrupt the fragility contrast.

4. **Sample size.** $n = 200$ per benchmark $\times$ model is a sample, not the full benchmark. Results generalize to the seed-42 matched subsets. Larger samples would narrow confidence intervals and potentially resolve cells where Bonferroni correction pushes borderline p-values above threshold (e.g., RAGTurk deepseek B-vs-A $p = 0.022$ before correction).

5. **Translation noise.** RAGTurk Arm C golds are machine-translated (single pass, deepseek-v4-flash). Translation error propagates into C scoring, potentially deflating C accuracy and creating artificial fragility signals. Human-translated golds would eliminate this confound.

6. **Model selection.** We test three models (gpt-5.6-luna, mimo-v2.5, deepseek-v4-flash). Other models deployed to Turkish users (e.g., GPT-4o, Claude, Gemini) may show different contamination profiles. Our findings are scope-limited to these three models.

7. **Benchmark selection.** We audit four benchmarks. Other Turkish evaluation datasets (e.g., Cetvel (Yıldız et al., 2025), Turkish SuperGLUE) may have different contamination profiles. The "no contamination detected" finding applies only to these four benchmarks under these probes.

8. **No corpus-overlap probe (M3).** The pre-registration described an M3 corpus-overlap modality using n-gram matching against training corpora. This was not executed in the current run due to cost and access constraints (requires corpus access or large-scale n-gram indexing). M3

would provide independent corroboration of M1/M2 signals.

9. **Run interruptions.** The run was interrupted and resumed three times due to process crashes. All resume logic is logged; no data was lost or duplicated. The run_start/run_end markers in the JSONL document the interruption history.

6 Conclusion

We present the first multi-probe contamination audit of the Turkish LLM evaluation ecosystem, covering four benchmarks × three models with a pre-registered threshold-sensitivity protocol. Under the adjudicated M1-DISCOUNT rule, no contamination is detected in any of the 12 benchmark × model cells. One cell reaches significance under RAW but is reclassified as threshold-dependent, demonstrating the importance of sensitivity analysis in contamination auditing.

The methodological contribution — the RAW + M1-DISCOUNT dual-reporting convention — generalizes beyond Turkish to any contamination audit that pools verbatim-recall signals with dynamic-arm signals. We release all data, code, and pre-registration artifacts for reproducibility and encourage the community to adopt multi-probe consensus auditing as standard practice for non-English benchmark validation.

Appendices

Appendix A: Pre-registration (verbatim)

The full pre-registration document is published alongside this paper. Key elements:

- **H0:** No statistically significant memorization signal detected by probes M1/M2 per benchmark × model cell.
- **Decision rule:** Flag only if ≥2/3 modalities indicate AND ≥1 dynamic arm $p < 0.05$ (Bonferroni).
- **Sample:** $n = 200$ seed-42 matched items per benchmark.
- **M1-DISCOUNT rule:** M1 struck from modality count; M2a/M2b must indicate independently.
- **Cost cap:** $2.00 hard abort.

Appendix B: Matcher documentation (RAGTurk fix 3)

The RAGTurk QA matcher uses a precision-first cascade: empty guard → letter-form → exact match → number guard → containment → paraphrase → long-descriptive. Measured on 106 hand-labeled pilot pairs: precision 1.000, recall 0.371, accuracy 0.792. The deliberate low recall means every RAGTurk accuracy is a conservative lower bound. A lenient sensitivity variant (no content-token guard) is logged per call and reported as a sensitivity column for RAGTurk only.

Appendix C: Spend log summary

Model	Calls	OK	F
GPT-5.6-luna	3,798	3,759	3
mimo-v2.5	3,798	3,798	
DeepSeek-v4-flash	5,597	5,597	
TOTAL	**13,193**	**13,154**	**3**

Table 8: Spend log summary for the full audit.

26 spend checkpoints logged; no cost cap breach. Total project spend including preparatory smoke/pilot runs: $0.89 (gpt-5.6-luna $0.43, deepseek-v4-flash $0.45, mimo-v2.5 uncosted).

References

References

Tameem Abbas et al. 2026. Obscuring data contamination through translation: Evidence from arabic corpora. In *arXiv preprint arXiv:2601.14994*.

Divyanshu Aggarwal et al. 2024. IndicMMLU-Pro: Benchmarking indic language models. In *arXiv preprint arXiv:2411.02457*.

Riyaz Ahmed et al. 2024. MathShikkha: A bangla mathematics education dataset and benchmark. In *arXiv preprint*.

Zeynep Barov et al. 2025. TUMLU: A unified multilingual benchmark for evaluating turkic languages. In *arXiv preprint arXiv:2502.11020*.

Beyond Benchmarks. 2025. Beyond benchmarks: Threats to evaluation in the llm era. In *arXiv preprint arXiv:2509.24210*.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, et al. 2020. Language models are few-shot learners. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Nicholas Carlini et al. 2021. Extracting training data from large language models. In *USENIX Security Symposium*.

ConStat. 2024. ConStat: Measuring non-generalizing performance in contaminated llm benchmarks. In *arXiv preprint arXiv:2405.16281*.

Contamination Multiplier. 2024. A multi-signal framework for benchmark contamination detection. In *arXiv preprint arXiv:2410.16186*.

Emine Dalğar et al. 2026. Halluverse-M3: A multilingual hallucination benchmark. In *arXiv preprint arXiv:2602.06920*.

Li et al. 2024. Why single perturbation tests fail: A multi-probe framework for benchmark contamination detection. In *arXiv preprint arXiv:2401.08501*.

GAOKAO-Eval. 2024. How do large language models fail chinese exam questions? a benchmark for chinese evaluation. In *arXiv preprint*.

Oguzhan Gencoglu et al. 2019. The ml reproducibility crisis. In *arXiv preprint arXiv:1902.06526*.

Xinyang Geng, Weiguang Jiao, et al. 2022. Detecting pretraining data from large language models. In *arXiv preprint arXiv:2210.10619*.

Tian Liang et al. 2024a. Evaluation of large language models: A survey. In *arXiv preprint arXiv:2402.05120*.

Weixin Liang, Xinyang Wang, et al. 2024b. Evaluating and mitigating data contamination for large language models. In *arXiv preprint arXiv:2410.16186*.

LINGOly. 2024. LINGOly: Benchmarking language models across 44 languages. In *arXiv preprint arXiv:2406.06196*.

Johannes Mattern, Fatemehsadat Mireshghallah, et al. 2023. Quantifying data contamination in code llms. In *arXiv preprint arXiv:2311.01964*.

METU NLP. 2026. RAGTurk: A turkish retrieval-augmented generation benchmark. In *arXiv preprint arXiv:2602.03652*.

Yoav Oren et al. 2023. Proving test set contamination in black box language models. In *arXiv preprint arXiv:2310.16789*.

Joelle Pineau et al. 2020. Improving reproducibility in machine learning research. In *arXiv preprint arXiv:2003.13991*.

José Francisco Sánchez Salido et al. 2025. None of the others: a general technique to distinguish reasoning from memorization in multiple-choice llm evaluation benchmarks. In *arXiv preprint arXiv:2502.12896*.

booktitle=arXiv preprint arXiv:2501.00593 Şimşek, Cihan and others, year=2025. TR-MMLU: A turkish benchmark for evaluating language models.

TR-MMLU Quality Audit. 2025. Setting standards for turkish llm evaluation. In *arXiv preprint*.

TurkEmbed. 2025. TurkEmbed4Retrieval: Turkish embedding model for retrieval. In *arXiv preprint arXiv:2511.07595*.

Jingfeng Yang, Xinyang Wang, Songyang Yang, et al. 2024. A survey of data contamination in large language models. In *arXiv preprint arXiv:2406.13236*.

Esra Yıldız et al. 2025. Evaluating benchmark quality for low-resource languages: A turkish case study. In *arXiv preprint arXiv:2504.09714*.

Haoyang Zhang et al. 2025. Test of time: Rethinking temporal signal of benchmark contamination. In *arXiv preprint arXiv:2509.00072*.